

Case 1 · Fluoroquinolone Tendinopathy Risk Amplified by Patient Factors

1 Patient Context	2 Reasoner (Initial)	3 Evidence Retrieval	4 Validator Judgment	5 Reasoner (Revised)
42F, left ankle pain 2 days after starting ciprofloxacin (Salmonella gastroenteritis) Risk factors ● Age > 40 ● Smoking 2 PPD × 25 yr ● Alcohol 2–3 beers/day ● BMI 30	Most likely cause of ankle pain? Initial answer B: Reactive arthritis (triggered by gastroenteritis) Plausible: temporal association with bacterial infection.	KG relations ciprofloxacin → contraindication → tendinitis Risk amplifiers: age >40, smoking, alcohol use Patient match Age > 40 ✓ Smoking ✓ Alcohol use ✓ Onset 2d after cipro → drug-induced timing	Claim B CONTRADICTED Onset 2d after cipro. All 3 risk amplifiers present. Drug-induced tendinopathy better explains the picture. Claim A SUPPORTED Adverse medication effect consistent with KG evidence + patient risk profile.	Revised answer A: Adverse medication effect (ciprofloxacin tendinopathy) ✓ Correct

Case 2 · Absolute Contraindication to tPA Identified from Patient-Specific Lab Value

1 Patient Context	2 Reasoner (Initial)	3 Evidence Retrieval	4 Validator Judgment	5 Reasoner (Revised)
71M, acute ischemic stroke (within thrombolytic window) History ● ITP (on prednisone) ● Plt 95,000/mm³ ● BP 175/105 mmHg	Which factor absolutely contraindicates IV tPA? Initial answer A: BP 175/105 mmHg (hypertension as contraindication) BP appeared most salient hazard for ischemic stroke.	KG relations low platelet count → increases risk → bleeding tPA eligibility threshold: plt < 100,000/mm³ Patient match Plt 95k < 100k ✓ ITP diagnosis ✓ BP 175/105: below SBP >185 threshold ✗	Claim A CONTRADICTED BP 175/105 is below absolute cutoff (SBP >185 / DBP >110). Not an absolute CI. Claim C SUPPORTED Plt 95k < 100k threshold. ITP + thrombocytopenia = absolute CI to tPA.	Revised answer C: Platelet count 95,000/mm³ (absolute CI to IV tPA) ✓ Correct